4.8 Specialized Asphalt Mixtures

4.8.1 Stone Matrix Asphalt Mix

Stone matrix asphalt (SMA) mixtures have been used in the United States since 1991. These mixtures have performed well in Europe for many years, and experience in the United States is similar. SMA is a rut-resistant, gap-graded HMA. The mix relies on stone-on-stone contact to provide its internal strength. The mix uses a high asphalt binder content of 6 to 7 percent or more. This high binder content provides improved durability. In most situations, the asphalt binder is a polymer-modified asphalt. Mineral filler and either mineral or cellulose fibers are added to form a stiff mortar and prevent draindown of the asphalt cement. The in-place air-voids are kept low—to less than 6 percent to insure that the mix is impermeable to water. SMA is routinely used in many states as the surface of choice for high-volume roadways and intersections. Figure 4.24 shows an SMA mix in Colorado.

Mix Design

The mix design procedure presented is based on AASHTO PP 41 (2004) *Designing Stone Matrix Asphalt (SMA)*. The mix design process consists of these five steps:

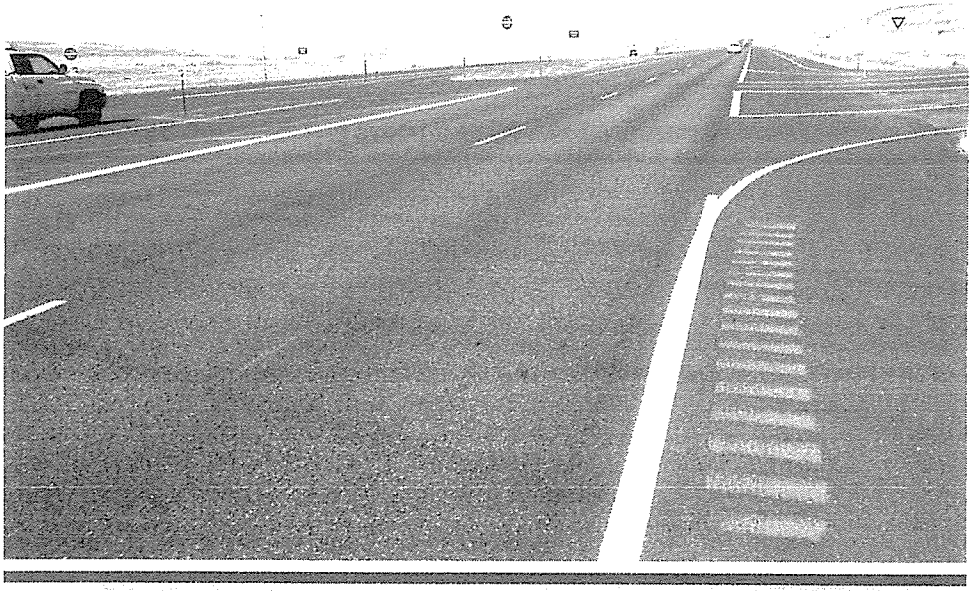


Figure 4.24 SMA Mix in Colorado

1. Select proper materials (aggregate, asphalt binder, mineral filler, and fibers).
2. Determine the aggregate gradation that yields stone-on-stone contact.
3. Ensure that the selected gradation provides the minimum required VMA.
4. Choose an asphalt binder content that provides the desired air-voids.
5. Evaluate the mix for moisture susceptibility.

Materials Selection

The materials required for an SMA mix include aggregates, asphalt binder, mineral filler, and a stabilizer, which is usually either a cellulose or a mineral fiber.

SMA mixtures require the use of a high-quality aggregate. AASHTO specifications require that 100 percent of the coarse aggregate have at least one crushed face and at least 90 percent must have two or more crushed faces. The strength of an SMA mixture is achieved through stone-on-stone contact. Therefore, most agencies have placed more stringent requirements on aggregate used in SMA. AASHTO requires a maximum Los Angeles Abrasion loss of 30 percent. But some states—because of a lack of high-quality aggregates—have built successfully performing projects with a less-stringent requirement.

The gradation for different SMA mixtures is shown in Table 4.9. Note that for the 19 millimeters, 12.5 millimeters, and 9.5 millimeters mixes, the mixtures have a low percentage passing the No. 4 sieve (4.75 millimeters) and a high percentage passing the No. 200 sieve. This is required to provide a mix with stone-on-stone contact and to provide a high VMA (generally 17 percent). The high percentage passing the No. 200 sieve (0.075 millimeter) is required to provide a stiff mix that will be rut resistant and to insure that draindown does not occur during construction. The high percentage of minus No. 200 material usually requires that as much as 5 to 7 percent commercial filler (for example, a limestone dust) is added to an SMA mix. Note that the design of an SMA mix is based on the volume of the materials (percent passing by volume) rather

Table 4.9
Typical Gradations for SMA Mixes (Percent Passing by Volume)

Sieve Size	Mixture Designation		
	19 mm	12.5 mm	9.5 mm
1 in (12.5 mm)	100		
¾ in (19.0 mm)	90–100	100	
½ in (12.5 mm)	50–88	90–100	100
⅜ in (9.5 mm)	25–60	50–80	70–95
No. 4 (4.75 mm)	20–28	20–35	30–50
No. 8 (2.36 mm)	16–24	16–24	20–30
No. 16 (1.18 mm)	–	–	21 max.
No. 30 (0.60 mm)	–	–	18 max.
No. 50 (0.30 mm)	–	–	15 max.
No. 200 (0.75 mm)	8–11	8–11	8–12

than by weight (as is done for Superpave, Marshall and Hveem mix designs). Therefore, it is necessary to convert the weight of materials into volumes. To accomplish this, the bulk specific gravity of the coarse aggregate, fine aggregate, and mineral filler must be known.

The asphalt binder for SMA mixtures has typically been a stiffer binder than is usually specified for dense-graded mixtures. They are usually polymer-modified mixes. The use of stiffer binders will tend to reduce the draindown of the asphalt binder during construction. A modified asphalt may also improve the high and low temperature properties of the SMA mix.

A stabilizer is typically used in the SMA mixture to prevent draindown of the asphalt binder during construction. Two types of fibers have been used: cellulose fibers and mineral fibers. Both types of fibers have been used successfully (Figures 4.25 and 4.26). The mineral fiber is usually heavier than the cellulose fibers and is therefore added at a higher percentage (by weight) so that a sufficient volume of fibers is added.

Selection of Desired Gradation

Once satisfactory materials have been identified, the optimum aggregate gradation and binder content must be determined. This is accomplished by first selecting an appropriate aggregate blend. This blended gradation should provide an aggregate skeleton with stone-on-stone contact and furnish a mixture that meets or exceeds the minimum voids in mineral aggregate. The condition of stone-on-stone contact is defined as the point at which the voids in the coarse aggregate fraction of the blend of the compacted SMA mixture (VCA_{mix}) is less than the VCA of the coarse aggregate fraction of the aggregate blend alone (VCA_{drc}) in the dry-rodded test (AASHTO T 19 or ASTM C 29).

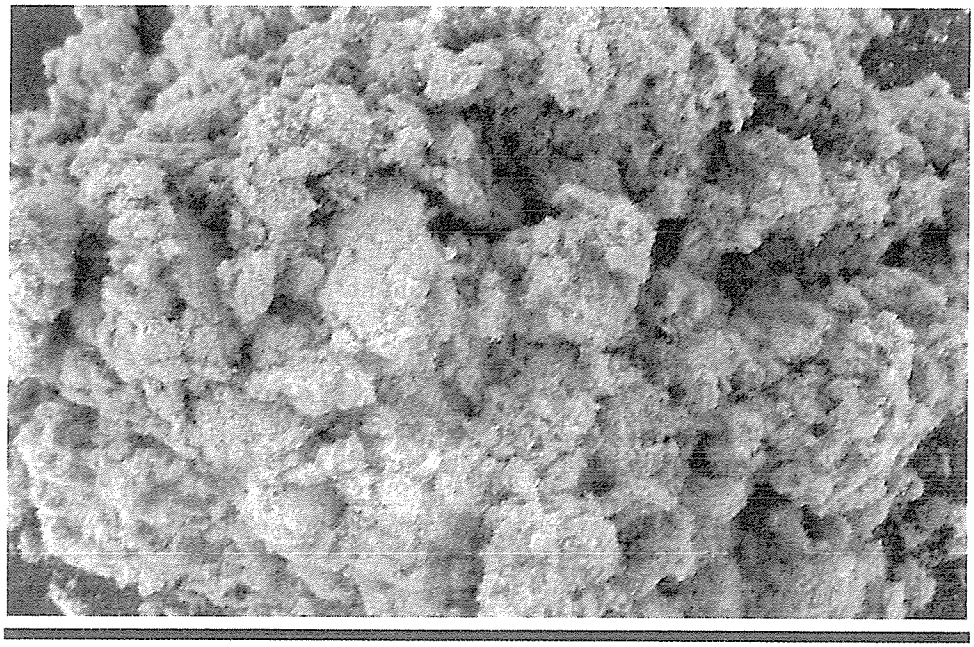


Figure 4.25 Cellulose Fiber

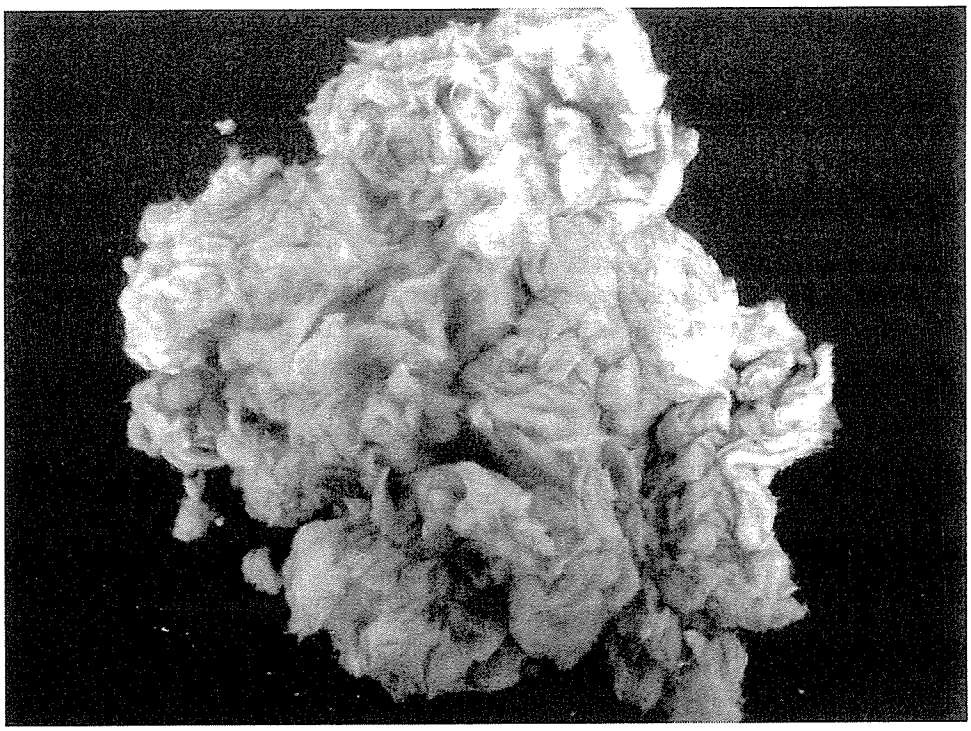


Figure 4.26 Mineral Fiber

The VCA of the coarse aggregate fraction is determined by compacting the aggregate with the dry-rodded technique according to AASHTO T 19, *Unit Weight and Voids in Aggregate.* The acronym VCA_{DRC} is used to indicate the voids in coarse aggregate of the coarse aggregate fraction in the dry-rodded condition. This value is used in the determination of stone-on-stone contact and is calculated as follows:

$$VCA_{DRC} = \frac{G_{ca}\gamma_w - \gamma_s}{G_{ca}\gamma_w} \times 100$$

where

VCA_{DRC} = voids in coarse aggregate in dry-rodded condition, percentage
γ_s = dry unit weight of coarse aggregate fraction in dry-rodded condition (kg/m³)
γ_w = unit weight of water (998 kg/m³)
G_{ca} = combined bulk specific gravity of coarse aggregate fraction

After the trial gradation samples have been compacted and allowed to cool, they are removed from the molds and tested to determine bulk specific gravity in accordance with AASHTO T 166. The volumetric properties of

each compacted sample are then determined. Following are the volumetric calculations of interest:

$$V_a, \% = 100 \times \left(1 - \frac{G_{mb}}{G_{mm}} \right)$$

$$VCA_{MIX}, \% = 100 - \left(\frac{G_{mb}}{G_{ca}} \times P_{CA} \right)$$

$$VMA, \% = 100 - \left(\frac{G_{mb}}{G_{sb}} \times P_s \right)$$

where

V_a = percentage of air-voids in compacted mixture
VCA_{mix} = voids in coarse aggregate fraction within compacted mixture
VMA = voids in mineral aggregate
G_{mb} = bulk specific gravity of compacted mixture,
G_{mm} = theoretical maximum specific gravity (AASHTO T 209 or ASTM D 2041)
P_s = percentage of aggregate in the total mixture
P_{CA} = percentage of coarse aggregate in the total mixture
G_{sb} = combined bulk specific gravity of the aggregates and
G_{ca} = combined bulk specific gravity of the coarse aggregate fraction

Of the trial blends evaluated, the one that meets or exceeds the minimum VMA requirement and has a VCA_{mix} less than VCA_{drc} should be selected as the design gradation. When an SMA mix has a VCA_{mix} less than VCA_{drc}, it has achieved stone-on-stone contact.

Once a trial blend has been selected as the design gradation, it may be necessary to raise or lower the asphalt binder content to obtain the proper amount of air-voids. Additional samples are prepared using the design gradation, and the binder content is varied.

The final chosen mix should meet the following requirements:

- Air-Voids at N_{design}: 4.0 percent
- Voids in Mineral Aggregate: 17 percent minimum
- Voids in Coarse Aggregate for Mix (VCA_{mix}): percentage should be less than VCA_{drc}
- Tensile Strength Ratio (AASHTO T 283): 70 percent minimum
- Draindown at Production Temperature: 0.30 percent maximum

Draindown

Due to the gap-graded nature of an SMA mixture, the possibility exists that the asphalt binder will drain off the aggregate during the construction process. To prevent this, a stabilizer is added to the mix. The draindown test is used to determine the correct amount of stabilizer needed for the mix. The test measures the potential for asphalt binder to drain from the coarse

aggregate structure while the mix is held at an elevated temperature. The test is performed in accordance with ASTM D 6390. To run this test, a sample is prepared in the laboratory (during mix design) or obtained from field production. The sample is placed in a wire basket (see Figure 4.27) that is put onto a suitable container of known mass (generally a paper plate). The sample, basket, and container are then placed into a forced-draft oven for one hour at or above the anticipated production temperature. At the end of one hour, the mass of asphalt binder draining from the sample that is retained in the container is determined and the amount of draindown calculated.

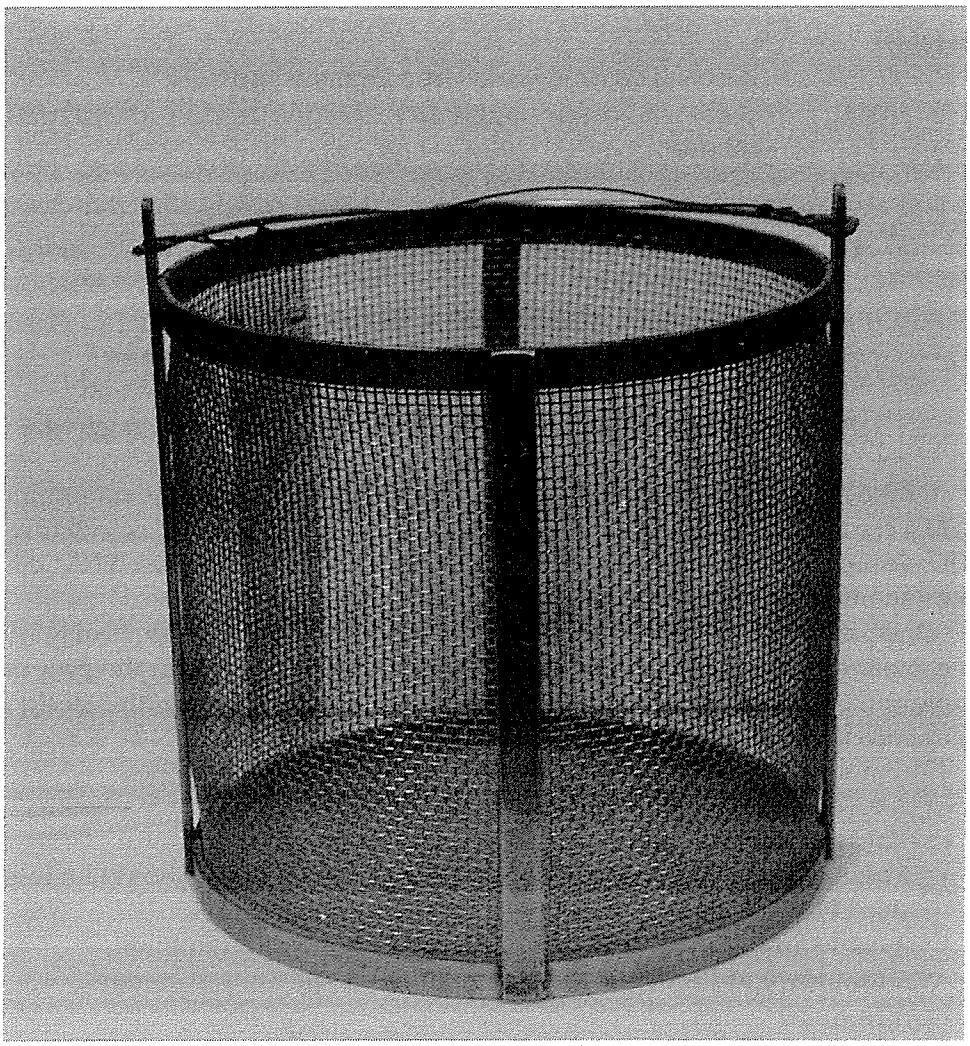


Figure 4.27 Draindown Basket

4.8.2 Open-Graded Friction Course

Description and Usage

Open-Graded Friction Courses (OGFC) provide an open mix with permeable voids that provide drainage on the surface of an HMA pavement under wet conditions. The water drains vertically through the OGFC to an impermeable, underlying layer and then laterally to the day-lighted edge of the OGFC. In addition to minimizing hydroplaning and providing high friction resistance on wet pavements, an OGFC layer will also provide the following advantages:

- Reduce splash and spray (see Figure 4.28)
- Reduce the potential for hydroplaning
- Enhance visibility of pavement markings
- Reduce nighttime surface glare in wet weather
- Reduce tire-pavement noise

While open-graded mixes provide many advantages, they have the following limitations:

- Snow- and ice-control procedures may need adjustment when they are used.
- They may ravel and shove when used at intersections, locations with heavy turning movements, and airfield ramp terminals.
- They may require special patching and rehabilitation techniques to maintain adequate drainage.

Open-graded mixes have been used throughout the United States since 1944. In many states they have been used extensively for many years. In the early 1970s, many states experienced problems with OGFC layers. These problems were raveling, delamination, loss of permeability after a few years of service, and stripping of the underlying layers. These problems have been addressed by one or more of the following: use of polymer-modified asphalt binders, relatively high asphalt binder contents

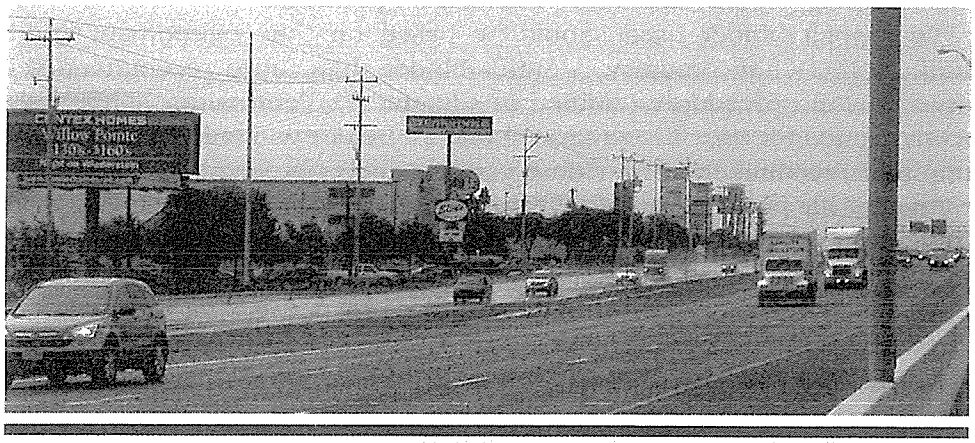


Figure 4.28 Comparison of Splash Spray Characteristics of an OGFC Surface (foreground) to a Conventional Surface (background)

(by the use of fibers and polymer modified asphalts), and by addressing the problems with moisture susceptibility in the underlying layers through the use of admixtures (lime or liquid antistrips).

Mix Design Characteristics and Procedures

The two published mix design procedures for the design of open-graded friction courses include the procedure outlined in the FHWA Technical Advisory T 5040.31 and a procedure based on research by the National Center for Asphalt Technology (NCAT). Most OGFC surface layers placed in the United States are designed by the specifying agency, using its own internal techniques and experience to accomplish the mix design. It is recommended that a designer check with the specifying agency about its procedure when designing OGFC mixes. The mix design consists of four steps:

1. Materials selection
2. Selection of the design gradation
3. Determination of the optimum asphalt content
4. Evaluation of moisture susceptibility

Materials Selection

The materials used for OGFC include aggregates, asphalt binders, and additives (such as fibers). The aggregates must be high-quality, polish-resistant aggregate materials. The NCAT procedure requires the coarse aggregate used in an OGFC to have a Los Angeles Abrasion value of less than 30 percent. Soft aggregates should be avoided due to the potential for excessive breakdown during mix production and compaction in the field. The aggregate should consist of 100 percent fractured faces to provide high internal friction. If crushed gravel is used, it should have at least 95 percent of the particles with two fractured faces and 100 percent with one fractured face. The fractured faces on gravels should be produced by mechanical fracturing of the rock. The percentage of flat and elongated particles should not exceed 5 to 20 percent corresponding to ratios of 5:1 and 3:1. The fine aggregate used in the mixture should have a fine aggregate angularity of at least 45 percent.

The asphalt binder used should be based on the environment and traffic levels of the roadway. A stiffer binder than those recommended based on the procedures outlined in Chapter 2, "Petroleum Asphalt" is recommended for use. It is suggested that a binder two grades stiffer than the grade recommended for dense-graded mixes be used and that they be made with polymer-modified asphalts. The addition of fiber may be desirable to reduce the amount of draindown during construction. Either a cellulose fiber is added at about 0.3 percent of total mix or a mineral fiber added at about 0.4 percent of total mix. The dosage rate for the fibers depends on the results of the draindown test.

Selection of Design Gradation

OGFC must have an aggregate structure that provides stone-on-stone contact to minimize rutting. As with the SMA mix, the condition of stone-on-stone contact is defined as the point at which the voids in the coarse aggregate fraction of the blend of the compacted OGFC mixture (VCA_{mix}) is less than the VCA of

the coarse aggregate (aggregate retained on the 4.75 millimeters sieve) fraction of the blend alone (VCA_{drc}) in the dry-rodded test (AASHTO T 19 or ASTM C 29). The equations for computing VCA_{mix} and VCA_{drc} for OGFC mixes are the same equations as those presented in the earlier section on SMA mixes.

Table 4.10 shows some typical gradations of OGFC mixes being used in the United States.

NCAT Procedure

The National Center for Asphalt Technology (NCAT) developed a procedure that is described in NCAT Report 00-01, *Design, Construction, and Performance of New Generation Open Graded Friction Courses*. The selection of the design gradation should consist of blending at least three trial blends ranging from the coarse to the fine side of the gradation band, with one falling in the middle of the band. For each trial gradation, the dry-rodded voids in the coarse aggregate fraction (VCA_{drc}) should be determined using AASHTO T 19 or ASTM C 29.

For each trial gradation, two specimens should be compacted at between 6.0 and 6.5 percent asphalt binder using 50 gyrations of a Superpave Gyratory Compactor. If a stabilizer is to be used in the final mix, it should be included in the trial mixes. If the VCA_{mix} is equal to or less than the VCA_{drc}, then stone-on-stone contact exists. If all the trial blends meet this requirement, choose a trial gradation that provides the highest air-voids (V_a) content.

To determine the VCA_{mix}, the designer must determine the bulk specific gravity of the mix (G_{mb}). The measurement of the bulk specific gravity (G_{mb}) of a compacted OGFC sample is a problem. This test is conducted by the water displacement method (AASHTO T 166 or ASTM D 2726) using saturated surface dry (SSD) specimens. The SSD method has been proven to work for dense-graded HMA mixtures. A problem with OGFC mixes is that they have large internal air-voids (15 to 25 percent) that are interconnected. During the bulk specific gravity testing with the SSD method, water will quickly infiltrate the sample. After the sample is removed from the water bath to obtain the SSD condition, the water will also drain from the sample

Table 4.10
Typical OGFC Gradations

Sieve Size	Percent Passing			
	NCAT	Florida	New Mexico	Nevada
¾" (19 mm)	100	100	100	100
½" (12.5 mm)	85–100	85–100	100	100
⅜" (9.5 mm)	35–60	55–70	90–100	90–100
No. 4 (4.75 mm)	10–25	15–25	25–55	35–55
No. 8 (2.36 mm)	5–10	5–10	–	–
No. 10 (mm)	–	–	0–12	–
No.16 (1.18 mm)	–	–	–	5–18
No. 40 (mm)	–	–	0–8	–
No. 200 (0.075 mm)	2–4	2–4	0–4	0–4

quickly. This draining of the water from the sample is a cause of errors in the water displacement method.

To overcome this problem, two approaches have been used to determine the bulk specific gravity of an OGFC. The first is to use dimensional analysis. In this technique, the weight of the sample is determined and the volume of the sample is obtained by carefully measuring the sample (height and diameter) and using that data to compute the volume. The bulk specific gravity is then determined by dividing the weight of the sample by the volume of the sample. Another approach is to use the CoreLok® vacuum-sealing device as discussed in Section 4.2.3.

Determination of Optimum Asphalt Content

Using the selected design gradation, the designer should prepare OGFC mixes at three binder contents. These samples should be subjected to the draindown test (ASTM D 6390). A maximum draindown of 0.3 percent by weight of mix is recommended for an OGFC mix.

Using the optimum asphalt binder content and gradation, the designer should conduct the Cantabro abrasion test on unaged and aged compacted samples. The test evaluates an OGFC material's ability to resist abrasion loss. A Los Angeles Abrasion machine is used to conduct the test. Each specimen is placed in the machine without the charge of steel balls. The machine is run for 300 revolutions at a speed of 30 to 33 rpm at room temperature. The percentage of abrasion loss is calculated using the following equation:

$$P = \left(\frac{P_1 - P_2}{P_1}\right) 100$$

where

P = percentage of abrasion loss and
P_1 = original weight of the sample and
P_2 = weight of the sample after abrasion testing

Resistance to abrasion will generally improve by increasing binder content or by using a stiffer PG binder grade of asphalt cement. In addition to the testing of the unaged mix, the design mix should be subjected to aging and then tested with the Cantabro test. OGFC mixes have a very high air-void content, which means that the asphalt binder in OGFC is prone to hardening at a faster rate than a dense-graded HMA. This may result in a reduction of the cohesive and adhesive strength of the binder, which could lead to raveling of the mix. Aging is accomplished by placing five compacted specimens in a forced-draft oven at 85°C (185°F) for 120 hours. The specimens are cooled to 25°C (77°F) and stored for four hours, after which the Cantabro test is conducted. The average abrasion losses on the five specimens should be less than 30 percent, with no specimens having an abrasion loss of more than 50 percent.

The resultant mix must meet the following requirements:

- Air-voids: 18 percent minimum (a higher percentage of air voids will result in a more permeable mix and a road surface that will produce lower noise levels)

- Voids in coarse aggregate for mix (VCA_{mix}): VCA_{mix} percent < VCA_{drc}.
- Abrasion loss (Cantabro test) on unaged specimens: 20 percent maximum
- Abrasion loss (Cantabro test) on aged specimens: 30 percent maximum
- Draindown at production temperature: 0.30 percent maximum

Evaluation of Moisture Susceptibility

The final mix should be evaluated for moisture susceptibility using the Modified Lottman method (AASHTO T 283 or ASTM D 4867) with five freeze/thaw cycles in lieu of one cycle. The retained tensile strength should be at least 80 percent.

FHWA Procedure

Determination of optimum asphalt content

The initial asphalt content is determined using the kerosene equivalent procedure (AASHTO T 270) to determine the surface capacity of the aggregate (K_c). The asphalt content is computed as follows:

$$P_b = (2K_c + 4.0) * \left(\frac{2.65}{GS_a}\right)$$

where

P_b = the design asphalt content
K_c = surface capacity of the aggregate
GS_a = apparent specific aggregate of the aggregate

Determination of optimum mixing temperature

A 1000-gram sample of aggregate is mixed with the proposed asphalt binder at the asphalt content determined above. It is mixed at a mix temperature corresponding to 800 centistokes using a temperature viscosity curve as discussed in Section 4.3. When the aggregate is completely coated, the mix should be transferred to a Pyrex glass plate (see Figure 4.34) and spread with a minimum of manipulation. The plate should be placed in an oven at the mixing temperature. Observe the bottom of the plate after 60 minutes. A slight puddle of asphalt cement at the points of contact between the aggregate and the glass plate is suitable and desirable after the 60-minute period.

Resistance to the effects of water

An immersion-compression test (AASHTO T 165 and T 167) (ASTM D 1075 and D 1074) is conducted on the designed mixture. The samples are prepared at the optimum mix temperature as determined above. Use a molding pressure of 2000 pounds per square inch. After a four-day immersion at 120°F (49°C), the Index of Retained Strength should not be less than 50 percent. Additives to promote adhesion will provide adequate retained strength when necessary.

Florida DOT Procedure

The initial asphalt content is determined based on the experience of the engineer or technician accomplishing the design. Factors to be considered are the gradation of the aggregate and the absorptive characteristics of the aggregate. A finer gradation will require more asphalt, and an absorptive aggregate will require additional asphalt binder.

After an initial asphalt content has been estimated, the Florida DOT mixes three 1200-gram samples—one at the estimated asphalt binder content, one at 0.5 percent above the estimated binder content, and one at 0.5 percent below the estimated binder content. These samples are mixed at 320°F (160°C). After the sample has been thoroughly mixed, it is placed in a pie plate. The pie plate is placed in an oven at 320°F (160°C) for one hour. It is taken out and cooled. The bottom of the pie plate is inspected to determine whether there is excessive draindown. As with the FHWA procedure, a slight puddle of asphalt cement at the points of contact between the aggregate and the glass plate is suitable and desirable after the 60-minute period.

4.8.3 Asphalt-Treated Permeable Base

General Description

An asphalt-treated permeable base (ATPB) is built as a base for an HMA pavement. The purpose of this layer is to intercept the water that enters through the pavement surface before it can reach the unbound layers, resulting in lowering the strength of those layers. The water enters the pavement through cracks in the surface and/or a poorly compacted and therefore permeable HMA layer. The layer should be designed to provide sufficient air-voids to make the layer permeable (approximately 10 percent). The performance and maintenance of these systems is discussed in more detail in Chapter 5 of this handbook.

Mix Design Characteristics and Procedures

The mix design of an ATPB is very similar to that used for an OGFC layer. The goal is to provide a free-draining layer with sufficient asphalt binder to cement the particles together and to prevent stripping of the asphalt binder from the aggregate.

Table 4.11 presents gradations being used by different agencies for an ATPB layer.

The asphalt binder can be either the standard grade of binder used by the agency (such as a 64–22) or a modified binder. The primary concern in the choice of asphalt binder is insuring the coating of the aggregate and minimizing the amount of draindown during construction. Coating is evaluated by visually observing the mix in the laboratory during mixing. Different binder contents should be evaluated. The draindown test discussed in the sections on SMA and OFGC can be used to evaluate the draindown characteristics of the mix. If the draindown exceeds 0.3 percent by weight of aggregate, steps should be taken to reduce the draindown, such as using fibers or a polymer-modified asphalt binder.

The air-voids in the mix can be evaluated in the same manner as with the OGFC mixes. The mix should be compacted using 50 gyrations with a SGC and the bulk specific gravity of the ATPB determined using the CoreLok® procedure.

Table 4.11
Typical Gradations Used for ATPB

Sieve Size	Indiana	Ontario	Caltrans	Florida
Asphalt content by weight of aggregate	3.1 to 4.7	4.2 to 5.4	2.0 to 3.0	2.0 to 3.0
1½ in (37.5 mm)	100	100	100	100
1 in (25 mm)	70–98	80–99	100	95–100
¾ in (19 mm)	50–85	68–90	90–100	–
½ in (12.5 mm)	28–62	54–76	35–65	25–60
¾ in (9.5 mm)	15–50	45–67	20–45	–
No. 4	10–20	35–45	0–10	0–10
No. 8	3–20	20–45	0–5	0–5
No. 16	1–15	12–36	–	–
No. 30	1–10	7–28	–	–
No. 50	0–7	3–18	–	–
No. 100	0–6	1–12	–	–
No. 200	0–4	0–5	0–2	–

4.9 Performance Testing of HMA

4.9.1 Introduction

Performance testing of HMA mixes is conducted to predict the performance of HMA prior to use on a transportation facility and to evaluate new materials and design tools to improve the performance of HMA pavements. HMA mixture testing can be divided into five general categories:

- Modulus—shear, dynamic and resilient modulus
- Creep—static and dynamic creep
- Flexural beam fatigue
- Empirical
- Wheel-tracking tests

4.9.2 Modulus Testing

The modulus of a material is used to calculate the expected deformation from a specified load. The modulus of elasticity is determined by applying increasing loads at a specified rate and measuring the resulting deformation. Loads are converted to stress by dividing by the cross-sectional area of the specimen section under load. Deformation is converted to strain by dividing the change in length (deformation) by the original (non-deformed) length. In shear testing, the shear modulus (G) of a material is determined by dividing the shear stress (τ) by the shear strain (γ). In axial testing, the modulus of elasticity (E) is determined by dividing the axial stress (σ) by the axial strain (ϵ).

Shear Modulus

The specimens used for shear and dynamic modulus testing are generally manufactured using the Superpave Gyratory Compactor (SGC). An HMA sample is obtained by mixing the materials in the laboratory or from a hot mix asphalt mixing facility. Mixtures prepared in the laboratory are subjected to short-term oven aging (STOA). For shear modulus testing a STOA of four hours at 275°F (135°C) is used. This is longer than the two hours used in Superpave for mix design. Stiffening of mixtures and binder frequently occurs in any HMA facility; therefore the STOA procedure is not used for plant-produced mixtures.

The shear modulus test is used to describe the viscoelastic behavior of an asphalt mix. It is described in AASHTO TP 7. A sinusoidal shear strain of 0.01 percent (peak-to-peak) is applied to the specimen at ten frequencies (0.01 to 10 Hz). The shear stress and strain are then used to calculate the complex shear modulus (G*) of the asphalt mixture. The loading rate for this test is selected based on what the material will experience under

field conditions. The test can be used to determine the effect of loading frequency (traffic speed) and temperature on an HMA mixture. The results of this test are used as an input into the Mechanistic Empirical Design Guide for HMA pavement thickness design.

Dynamic (Axial) Modulus

The dynamic modulus, sometimes called complex modulus, is determined by applying sinusoidal (haversine) vertical loads to the specimen at different frequencies and measuring the recoverable vertical strain. The test procedure for determining the dynamic modulus is provided in AASHTO TP 2 or ASTM D 3497. The axial stress and strain are used to calculate the dynamic modulus. The dynamic modulus is calculated by dividing the repeated stress by the repeated strain. The difference between this test and the shear modulus test is the type of loading (axial versus shear loading) as well as the fact that the dynamic modulus test is stress-controlled while the shear modulus test is strain-controlled. This test is used to describe the viscoelastic behavior of asphalt materials under axial loading. The results of this test are used to evaluate the rutting resistance of an HMA mixture.

Resilient Modulus (RM) Test

The resilient modulus test is a repeated load indirect tension test. The test method is described in ASTM D 4123. See Figure 4.29 for a photo of the test set up. The specimen used for the resilient modulus test is a 2½ inches

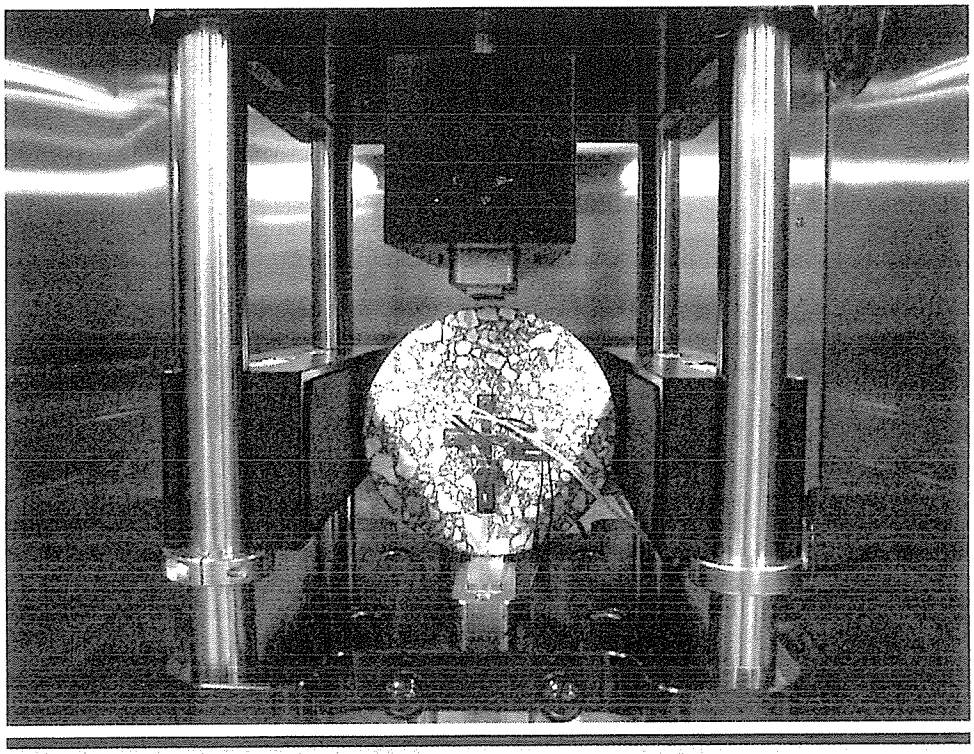


Figure 4.29 Resilient Modulus Test Set-up

(62.5 millimeters) by 4 inches (100 millimeters) specimen. The HMA specimens are normally loaded to a stress level between 5 and 20 percent of indirect tensile strength. A repeated load with constant frequency is applied to the specimen. The strain is measured with LVDT (Linear Variable Differential Transducer) and used along with the applied stress to compute the resilient modulus of the mixture. The results of this test are used as input into layered elastic thickness design procedures and are used to determine the structural coefficient for use in the Asphalt Institute and AASHTO thickness design procedures.

4.9.3 Creep Testing

Creep testing is a simple version of a controlled stress test. There are two types of creep tests—static creep and dynamic creep. In the static creep test a constant axial load is applied to the test specimen throughout the duration of the test. In the dynamic creep test, a large number of repeated axial load applications are applied to the specimen. It is called "dynamic" because the specimen is being continually loaded and unloaded throughout the duration of the test.

Static creep

In the static creep test an axial load is rapidly applied (within 0.05 sec) to the test specimen until a desired axial stress is achieved (usually between 69 and 207 kiloPascals). This axial stress is then maintained until the total axial strain reaches approximately 2 percent (20,000 microstrain) or tertiary flow occurs in the test specimen. The general shape of the axial strain curve is shown in Figure 4.30.

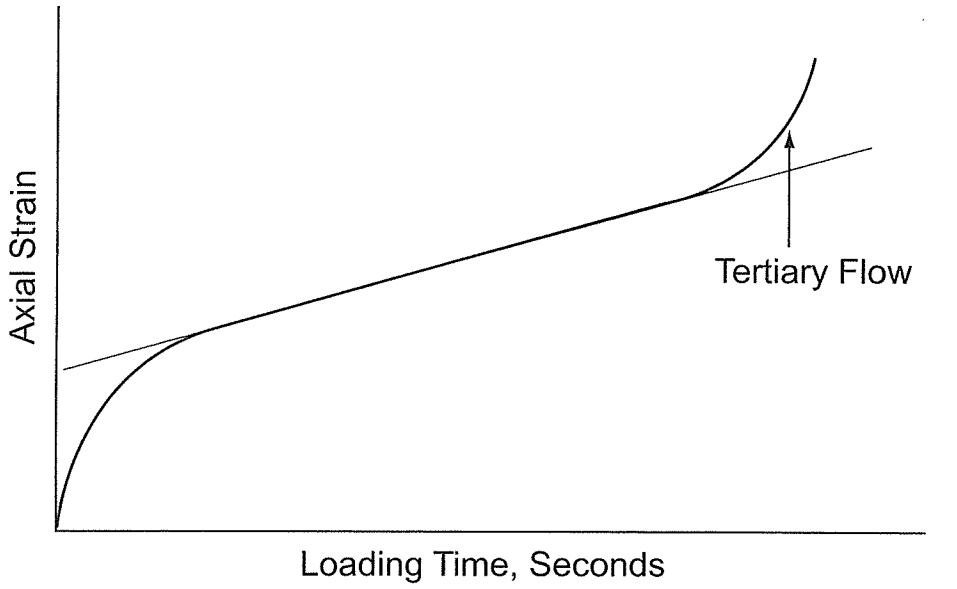


Figure 4.30 Axial Static Creep Test—Strain Response

At any time during the test, the applied stress can be divided by the observed strain to arrive at a creep modulus or stiffness. Because the strain changes with time, so does the creep modulus. In creep testing, many engineers prefer to compute creep compliance rather than creep stiffness. Creep compliance can be portrayed as the inverse of creep stiffness (although it is not strictly correct, theoretically). In other words, creep compliance is the strain at any time during the test per unit of stress. At the conclusion of a creep test, the test specimen will partially rebound to its original shape. For many materials, however, some permanent deformation remains in the test specimen.

The rate of change of creep compliance versus loading time is then calculated and plotted as shown in Figure 4.31. The minimum point of the curve occurs where the rate of creep compliance changes as a function of time is zero. This point is identified as the flow time and is the time when shear deformation occurs under constant volume. Higher values of flow time (FT) from the axial static creep test are thought to be related to better rutting resistance in the HMA mixture.

Dynamic Creep Test

Dynamic creep testing can also be conducted using repeated axial stress loading cycles and measuring the resulting axial strain. The concepts of repeated load axial (dynamic creep) testing are similar to the repeated shear test at constant height. A specified axial stress (usually 69–207 kilopascal) is applied to a test specimen for 0.1 seconds and released to a rest period of 0.9 seconds. Each loading cycle is 1 second (1 Hertz). In the standard test, loading cycles are applied for 10,000 cycles (approximately 2.75 hours) or until excessive tertiary deformation is reached in the test specimen. As with

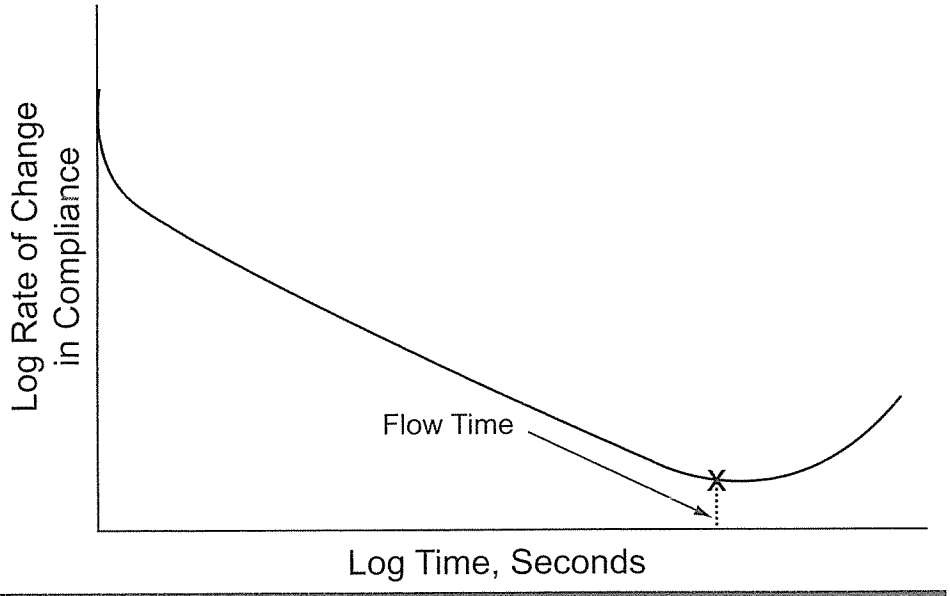


Figure 4.31 Rate of Change in Compliance as a Function of Loading Time

the repeated shear test, the axial strain is calculated and the permanent axial strain determined at the end of each loading cycle (ϵ_n).

The rate of change of axial strain versus loading time is calculated and plotted as shown in Figure 4.32. The minimum point of the curve occurs where the rate of change in axial strain as a function of time is zero. This point is identified as the flow number (FN) and is postulated to be the number of load repetitions at which shear deformation (under constant volume) occurs. Higher values of flow number (FN) from the repeated load axial (dynamic creep) test are thought to be related to better rutting resistance in the HMA mixture.

4.9.4 Flexural Fatigue Test

The flexural fatigue test is conducted to evaluate the fatigue characteristics of an HMA mixture. Fatigue cracking of pavement is considered to be more a structural problem than simply a material problem. Several external factors influence the fatigue cracking in pavements, such as poor subgrade drainage, time of placement, and method of compaction and placement of the asphalt mix.

The specimens for this test are 2.5 inches (62.5 millimeters) by 2 inches (50 millimeters) by 15 inches (375 millimeters) beams. The test is conducted in accordance to the procedures in AASHTO T 321. In this method, repeated haversine loads (0.1 second load and 0.4 second unload) are applied at the third points of the specimen. The beam fatigue test can be conducted in controlled stress or controlled strain mode. A photo of the apparatus is presented in Figure 4.33.

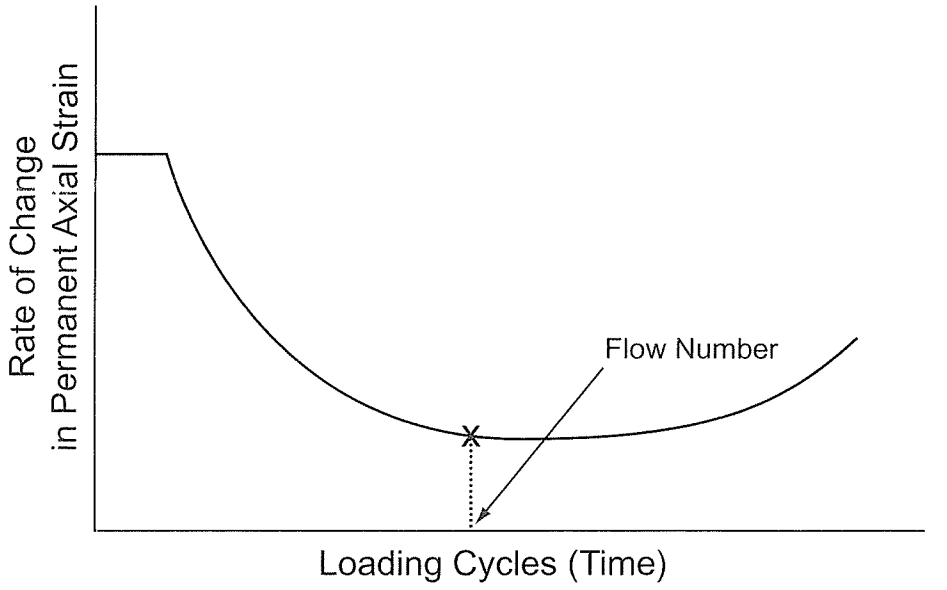


Figure 4.32 Rate of Change in Axial Strain as a Function of Loading Cycles (Time)

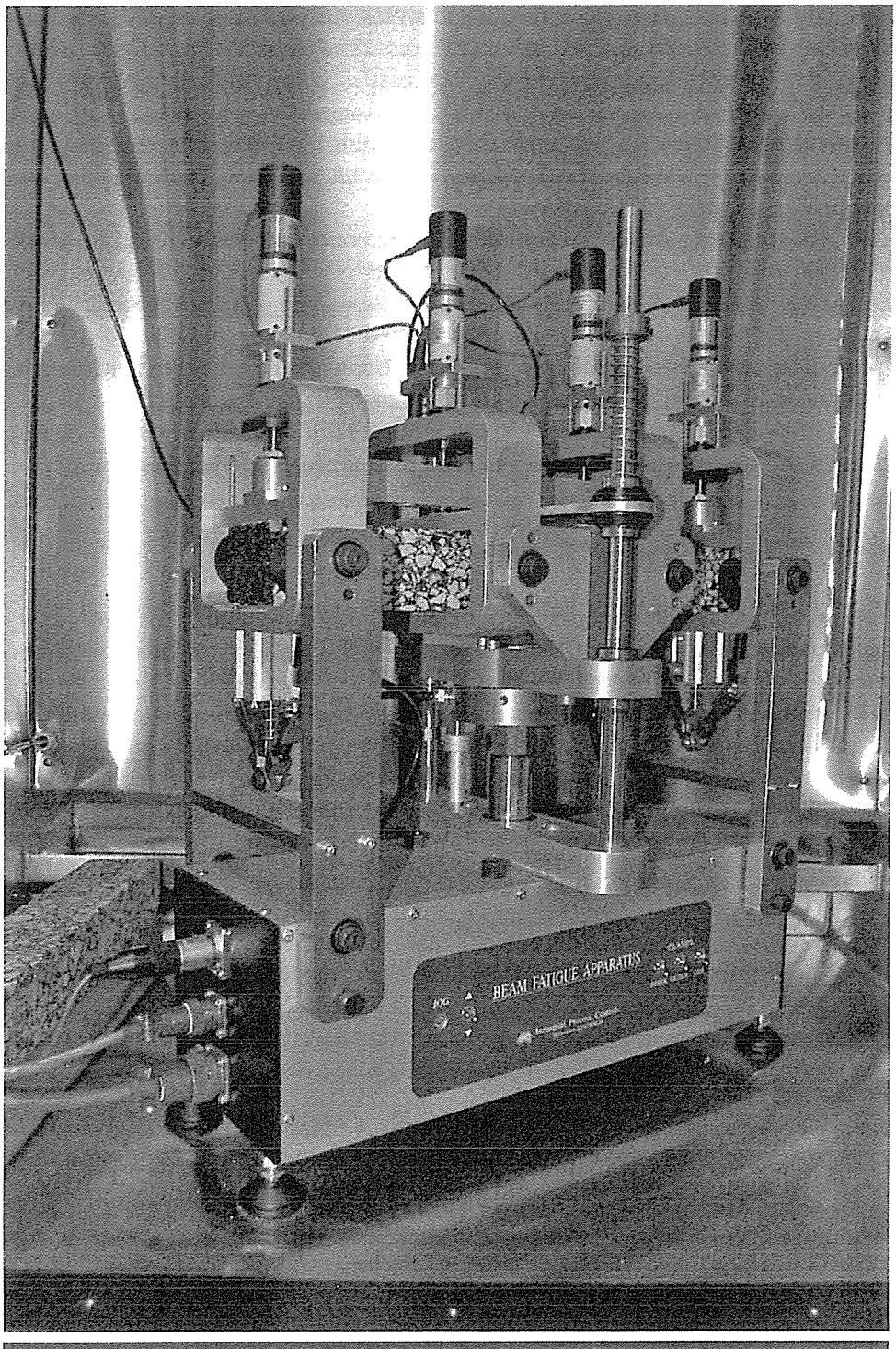


Figure 4.33 Beam Fatigue Test Apparatus

In the constant stress method, load cycles are applied until either the specimen fails to take more load or it breaks. The number of cycles required for a specimen is logged. Similarly, the test is conducted on several samples with different magnitudes of load and logged in a log-log plot. In the constant stress test, more samples with higher stiffness perform better than the samples with lower stiffness. The test results from this test are applicable to thick HMA pavements.

In the constant strain beam fatigue test, constant strain is maintained in a sample, but the magnitude of the load is allowed to decrease with increasing load cycles. In this method, the samples do not break during the test. Failure is defined when the applied load is reached at a certain percentage of the original applied load. Usually, failure is defined as 50 percent loss in initial beam stiffness. The test results are applicable to thin pavements—less than 2 inches (50 millimeters) thick. A typical plot of fatigue data in the constant strain beam fatigue test is presented in Figure 4.34.

4.9.5 Empirical Tests

Empirical tests have been used to relate the performance of an HMA pavement to a standardized test. The results from the test do not provide fundamental engineering properties that can be used in calculations of stresses and strains in an HMA pavement. Rather, they have shown through field performance to be indicative of performance. The three most commonly used empirical tests for HMA pavements are the Marshall test, the Hveem test, and the indirect tensile test; these were described earlier in this chapter.

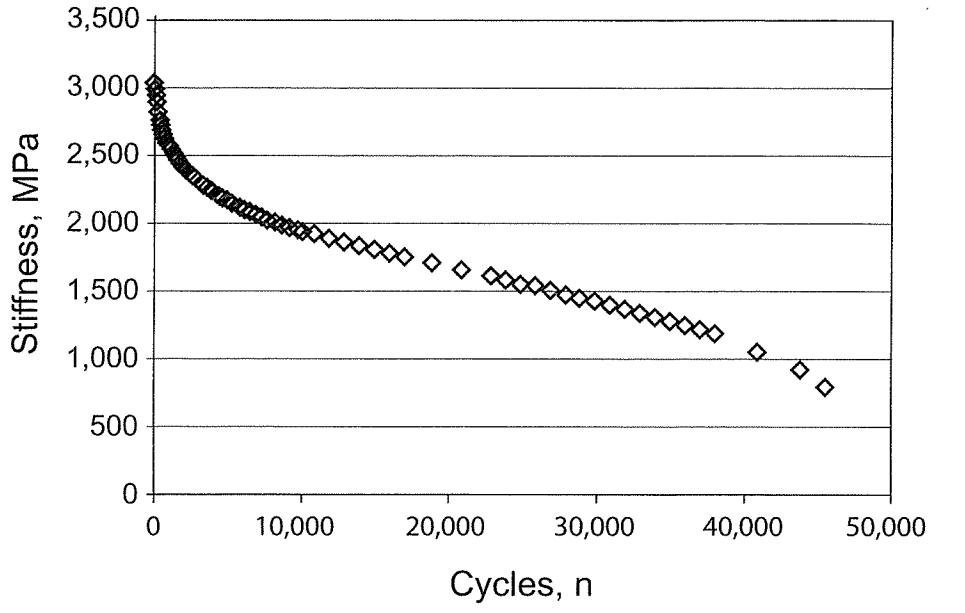


Figure 4.34 Typical Plot of Fatigue Data in Constant Strain Beam Fatigue Test

4.9.6 Wheel-Tracking Tests

Hamburg Wheel-Tracking Device (HWTD)

This machine was developed by the Germans in the 1970s. The test procedure is AASHTO T 324. The HWTD measures the combined effects of rutting and moisture on an HMA mixture. A steel wheel rolls across the surface of the HMA core, which is immersed in hot water. The steel wheel moves through a crank connected to a flywheel. The steel wheel has a diameter of 8 inches (204 millimeters) and is 1.85 inches (47 millimeters) wide with a 154-pound load applied. The testing is done with the samples in a hot water bath at 50°C. The test is run for 10,000 cycles (20,000 passes) or until a deformation of 0.787 inch (20 millimeters) is reached. The machine is shown in Figure 4.35.

Figure 4.36 is a sample of the typical data obtained from the HWTD. Three points are identified on the curve: the creep slope, the stripping slope, and the stripping inflection point. The creep slope occurs before the onset of stripping and is the inverse of the rate of deformation in the linear range of the deformation curve. This portion of the curve is where rutting will occur due to consolidation and plastic flow. The stripping slope is related to the severity of the damage due to moisture. It is the inverse of the rate of deformation in the linear region after the onset of the stripping to the end of the test. The stripping inflection point is related to the resistance of the HMA tested to the effects of the moisture. This point is determined at the number of passes at which the creep slope and the stripping slope intersect.

In its executive summary on performance testing of HMA mixes, NCAT suggested rut-depth criteria of less than 0.393 inch (10 millimeters) after 10,000 cycles (20,000 passes) for conventional dense-graded HMA mixes.

Asphalt Pavement Analyzer (APA)

The Asphalt Pavement Analyzer (APA) is a modification of the Georgia Loaded Wheel Tester (GLWT), which was developed in a cooperative study of the Georgia Department of Transportation and the Georgia Institute of Technology in 1985. It has been used primarily to evaluate the rutting resistance of an HMA mixture. Studies have been conducted to evaluate the use of the equipment for evaluating fatigue and moisture resistance of HMA mixtures. The APA is shown in Figure 4.37.

A loaded wheel is placed on a pressurized hose, which sits on the test specimens, and then tracked back and forth to induce rutting. Generally, the specimens are tested to 8,000 cycles. Samples can be tested dry or while submerged in water.

The test specimens for the APA can be either beam or cylindrical specimens. The beam samples are generally compacted using the Asphalt Vibratory Compactor (Figure 4.38). The cylindrical specimens are made with the Superpave Gyratory Compactor. The beams are compacted to 7 percent air-voids, and the cylindrical specimens are compacted to both 4 and 7 percent air-voids. Cores from an in-service roadway can also be tested.

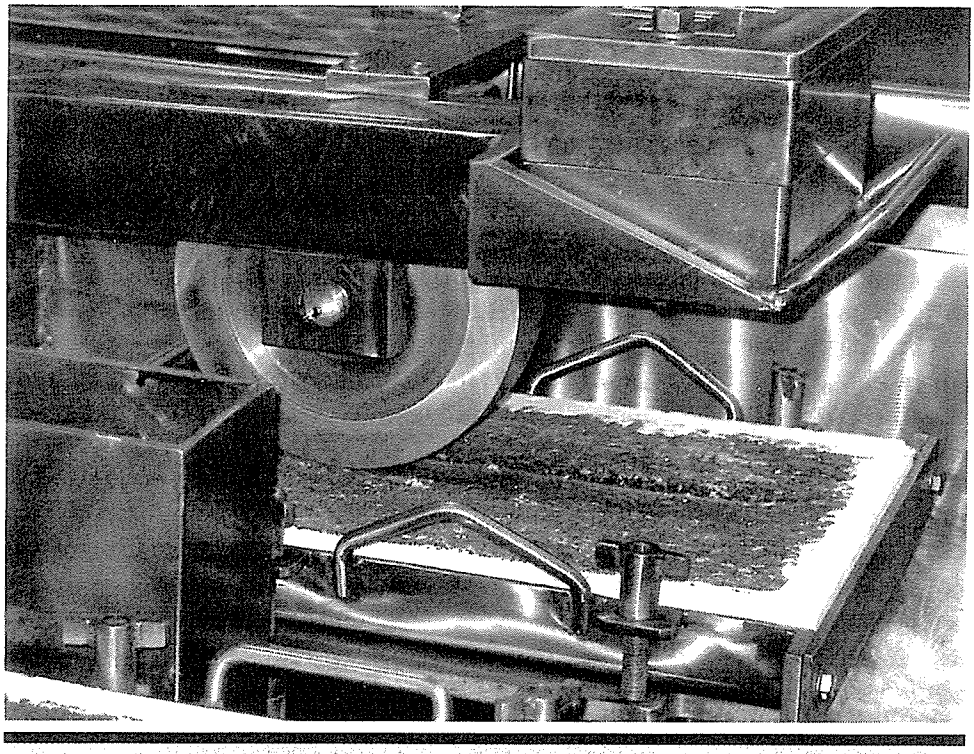


Figure 4.35 Hamburg Wheel-Tracking Device

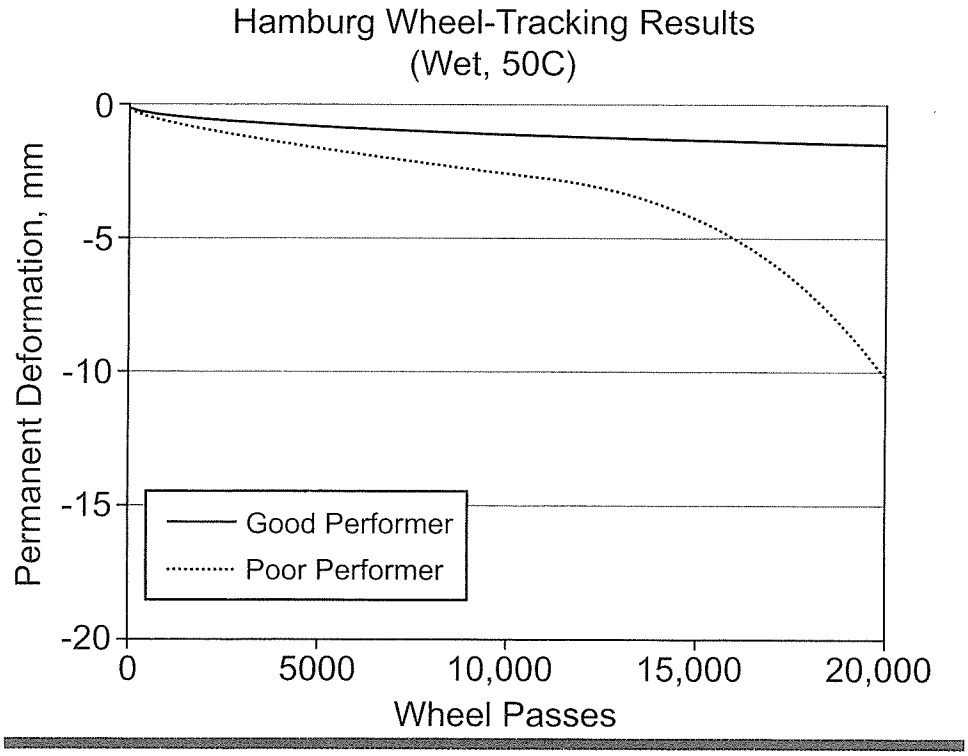


Figure 4.36 Typical Results from a Hamburg Wheel-Tracking Device

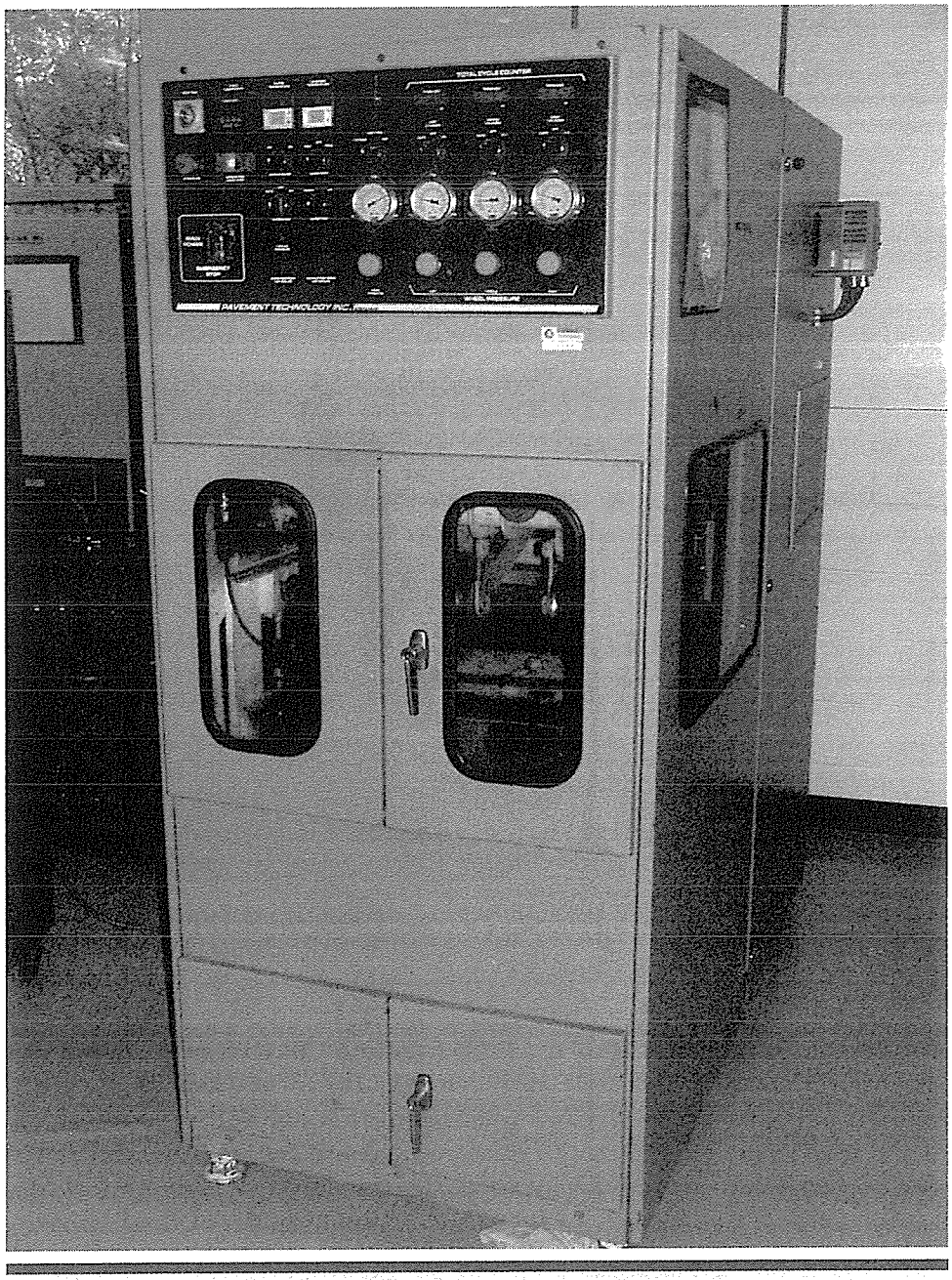


Figure 4.37 Asphalt Pavement Analyzer

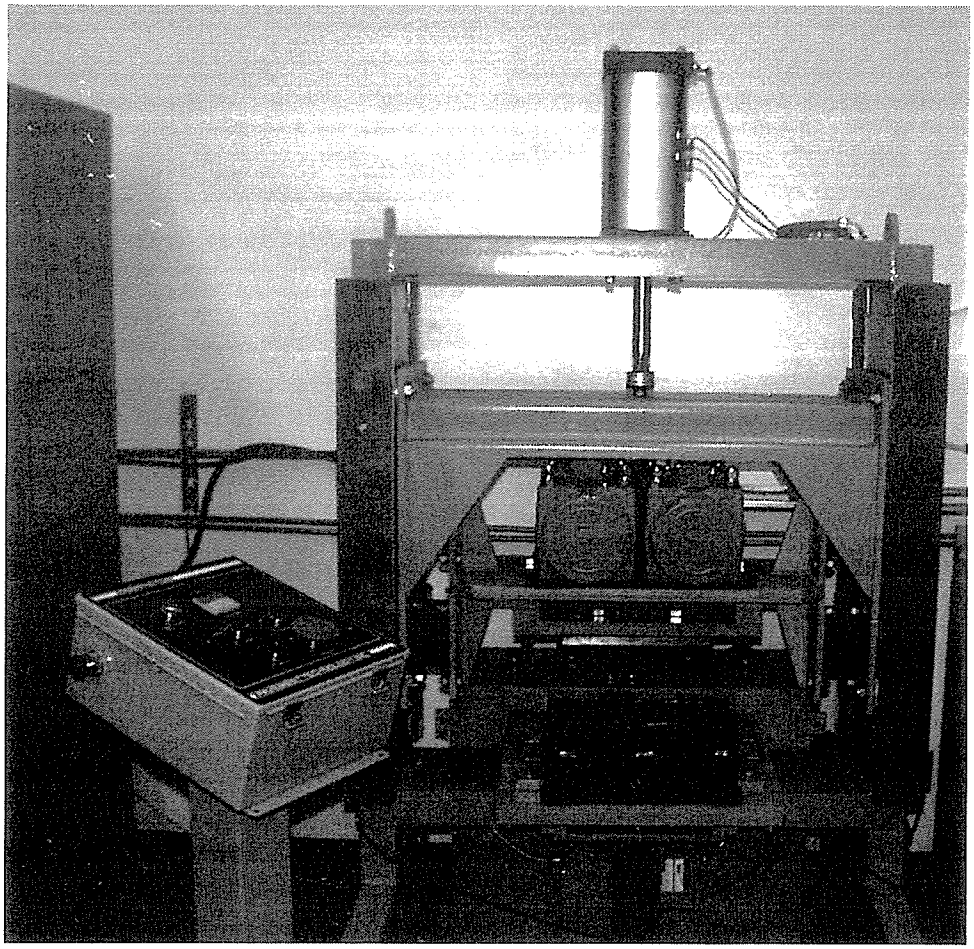


Figure 4.38 Asphalt Vibratory Compactor

The samples are tested at or near the maximum pavement temperatures (generally defined as the high temperature grade for the asphalt binder to be used in region where the HMA pavement is being built).

The wheel load is 100 pounds (445 Newtons) and the hose pressure is 100 pounds per square inch (690 kiloPascals). Figure 4.39 shows the typical APA rut-test results. It indicates that specimens deform rapidly at the beginning of the test. The amount of permanent deformation per cycle decreases and becomes quite steady after a certain number of load cycles.

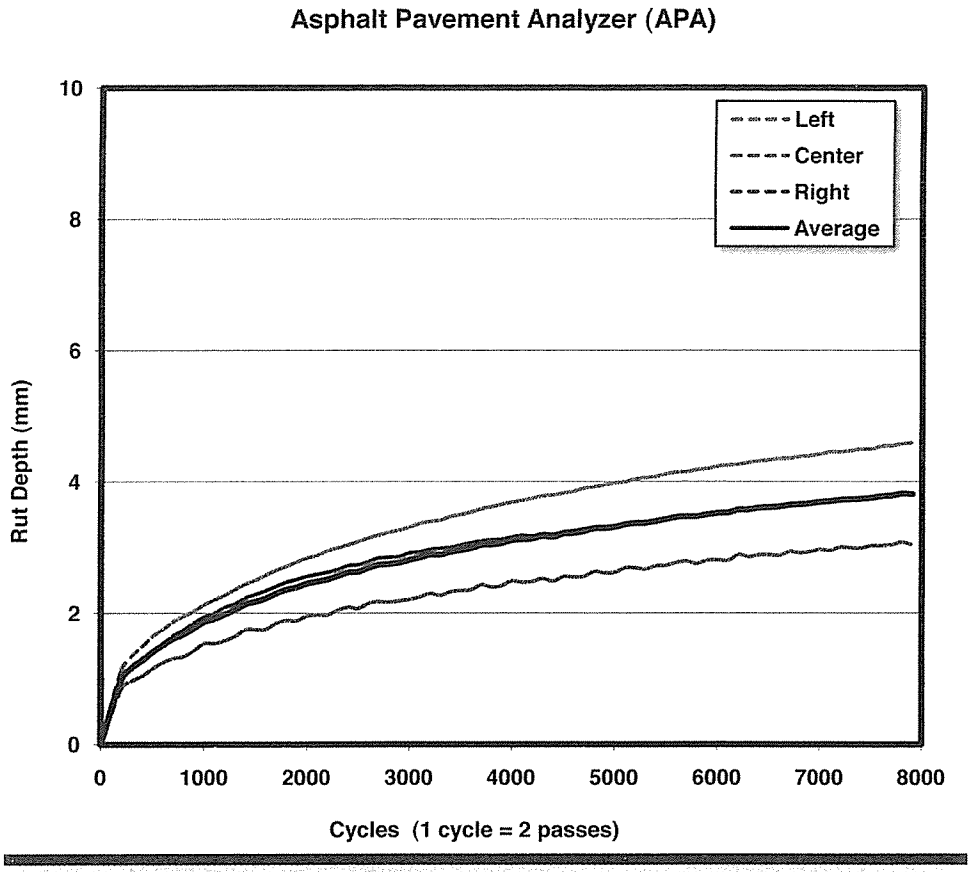


Figure 4.39 Typical APA Test Results